

# Salesforce Apex Hours



## Automate the App Lifecycle with CumulusCI

Speaker

David Reed, Salesforce.org

Date

Venue/Link

Online



#SalesforceApexHours



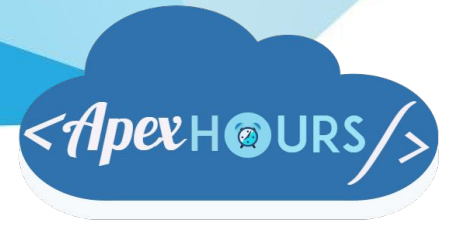
[www.apexhours.com](http://www.apexhours.com)

## Amit Chaudhary (Salesforce MVP)

- Active on Salesforce Developer Community
- Blogging at <http://amitsalesforce.blogspot.in/>
- Co-Organizer of – FarmingtonHillsSFDCDug
- Founder of [www.ApexHours.com](http://www.ApexHours.com)
- Follow us @Amit\_SFDC or @ApexHours



# Our Speaker



## David Reed

- Lead Member of Technical Staff, Release Engineering Salesforce.org
- 9x certified
- Moderator, Salesforce Stack Exchange
- @aorisdual / d.reed@salesforce.com
- www.ktema.org



# Agenda



The trail to production and the org problem

The Salesforce.org solution: portable automation with CumulusCI

Live demo

Metecho: Portable automation for admins

Completing the picture with MetaCI and MetaDeploy

Takeaways



# Forward-Looking Statement



Statement under the Private Securities Litigation Reform Act of 1995:

This presentation contains forward-looking statements about the company's financial and operating results, which may include expected GAAP and non-GAAP financial and other operating and non-operating results, including revenue, net income, diluted earnings per share, operating cash flow growth, operating margin improvement, expected revenue growth, expected current remaining performance obligation growth, expected tax rates, the one-time accounting non-cash charge that was incurred in connection with the Salesforce.org combination; stock-based compensation expenses, amortization of purchased intangibles, shares outstanding, market growth and sustainability goals. The achievement or success of the matters covered by such forward-looking statements involves risks, uncertainties and assumptions. If any such risks or uncertainties materialize or if any of the assumptions prove incorrect, the company's results could differ materially from the results expressed or implied by the forward-looking statements we make.

The risks and uncertainties referred to above include -- but are not limited to -- risks associated with the effect of general economic and market conditions; the impact of geopolitical events; the impact of foreign currency exchange rate and interest rate fluctuations on our results; our business strategy and our plan to build our business, including our strategy to be the leading provider of enterprise cloud computing applications and platforms; the pace of change and innovation in enterprise cloud computing services; the seasonal nature of our sales cycles; the competitive nature of the market in which we participate; our international expansion strategy; the demands on our personnel and infrastructure resulting from significant growth in our customer base and operations, including as a result of acquisitions; our service performance and security, including the resources and costs required to avoid unanticipated downtime and prevent, detect and remediate potential security breaches; the expenses associated with new data centers and third-party infrastructure providers; additional data center capacity; real estate and office facilities space; our operating results and cash flows; new services and product features, including any efforts to expand our services beyond the CRM market; our strategy of acquiring or making investments in complementary businesses, joint ventures, services, technologies and intellectual property rights; the performance and fair value of our investments in complementary businesses through our strategic investment portfolio; our ability to realize the benefits from strategic partnerships, joint ventures and investments; the impact of future gains or losses from our strategic investment portfolio, including gains or losses from overall market conditions that may affect the publicly traded companies within the company's strategic investment portfolio; our ability to execute our business plans; our ability to successfully integrate acquired businesses and technologies, including delays related to the integration of Tableau due to regulatory review by the United Kingdom Competition and Markets Authority; our ability to continue to grow unearned revenue and remaining performance obligation; our ability to protect our intellectual property rights; our ability to develop our brands; our reliance on third-party hardware, software and platform providers; our dependency on the development and maintenance of the infrastructure of the Internet; the effect of evolving domestic and foreign government regulations, including those related to the provision of services on the Internet, those related to accessing the Internet, and those addressing data privacy, cross-border data transfers and import and export controls; the valuation of our deferred tax assets and the release of related valuation allowances; the potential availability of additional tax assets in the future; the impact of new accounting pronouncements and tax laws; uncertainties affecting our ability to estimate our tax rate; the impact of expensing stock options and other equity awards; the sufficiency of our capital resources; factors related to our outstanding debt, revolving credit facility, term loan and loan associated with 50 Fremont; compliance with our debt covenants and lease obligations; current and potential litigation involving us; and the impact of climate change.

Further information on these and other factors that could affect the company's financial results is included in the reports on Forms 10-K, 10-Q and 8-K and in other filings it makes with the Securities and Exchange Commission from time to time. These documents are available on the SEC Filings section of the Investor Information section of the company's website at [www.salesforce.com/investor](http://www.salesforce.com/investor).

Salesforce.com, inc. assumes no obligation and does not intend to update these forward-looking statements, except as required by law.



# Let's Get Cost Out of the Way

This is not a sales pitch



CumulusCI, MetaCI, Metecho, and MetaDeploy are 100% free and open source.

Salesforce licenses and server infrastructure do cost money.

- You'll still pay Heroku, CircleCI, Jenkins, et al.  
for infrastructure if you wish to use continuous integration

A wooden signpost with a brown wooden board and a dark brown wooden post. The sign is set against a background of a green tree and a blue sky. The text on the sign is white and reads "Support the Salesforce open source community!".

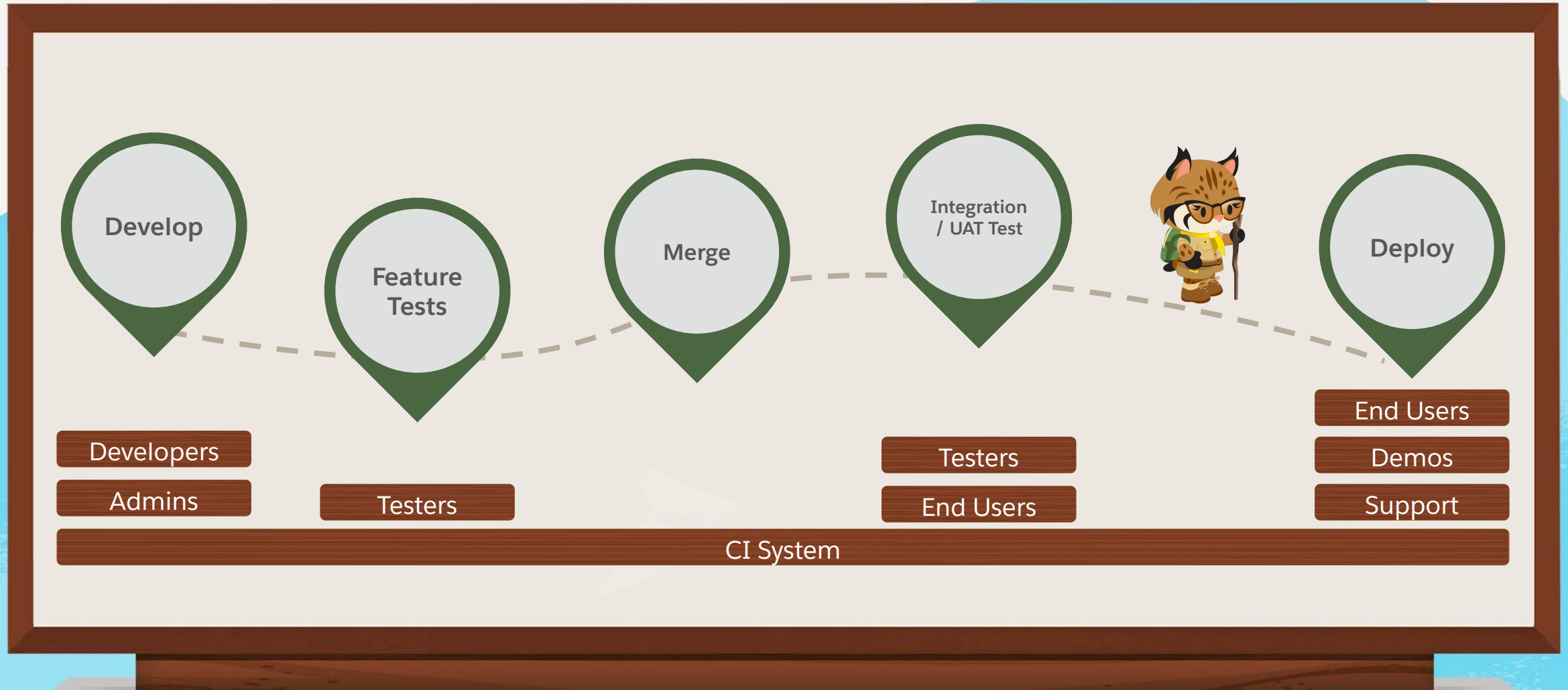
Support the Salesforce  
open source  
community!



# The Trail to Production

Exploring the app lifecycle

# The trail to production





# Where do these participants do their work?



Continuous integration must be automated.

Most CI systems do very similar steps:

- Create a Salesforce org.
- Configure the Salesforce org to meet the needs of the application.
- Deploy the application.
- Execute tests (Apex unit tests, Jest tests, Robot browser tests...)
- Report results

In many organizations, these steps are so complex that they end up using a single, persistent org.

Developers, admins, testers, support, product managers, and other stakeholders usually must create environments manually.

Each user has to:

- Create a Salesforce org.
- Configure the Salesforce org to meet the needs of the application.
- Deploy the application.

In many organizations, these steps are so expensive that users must make do with shared, out-of-date, or inconsistent sandboxes.

# The org problem

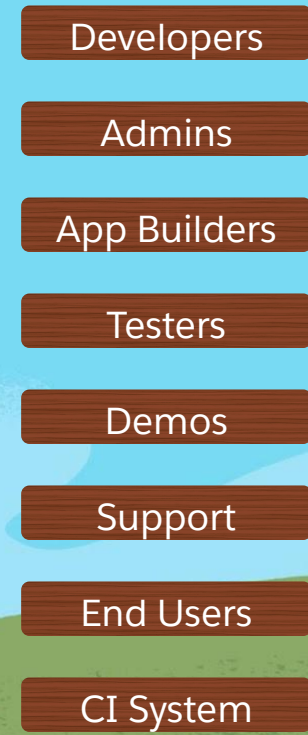
All of these roles and systems need Salesforce orgs to work in.

Building new orgs is hard.

- Traditional tools only do metadata setup.
- What about configuration? Data? Dependency management?
- What about multiple org shapes for different roles?
- CI automation is usually not “portable” (CI only).

Managing persistent orgs’ state is hard.

- Challenging to fully synchronize with source control
- Weak change management
- Users conflicting with one another

A vertical stack of eight brown rectangular buttons with white text, set against a background of a blue sky and green hills. The buttons are labeled: Developers, Admins, App Builders, Testers, Demos, Support, End Users, and CI System.

Developers

Admins

App Builders

Testers

Demos

Support

End Users

CI System

# Some consequences of the org problem



## Loss of velocity

- Teams can't work at capacity because of poor infrastructure

## Loss of confidence

- Test results aren't reliable without repeatable setup

## Difficulty implementing modern best practices

- Source control
- Salesforce DX

## Limited alternatives

- Automation solutions are too limited, too expensive, or require specialization.



# The Salesforce.org Solution: Portable Automation with CumulusCI

Portable automation for every role,  
everywhere on the trail

# Basic framework



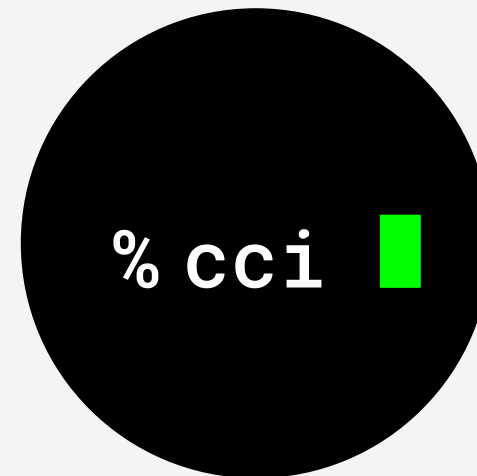
Version control as  
source of truth.



Salesforce DX scratch  
orgs as location of work.

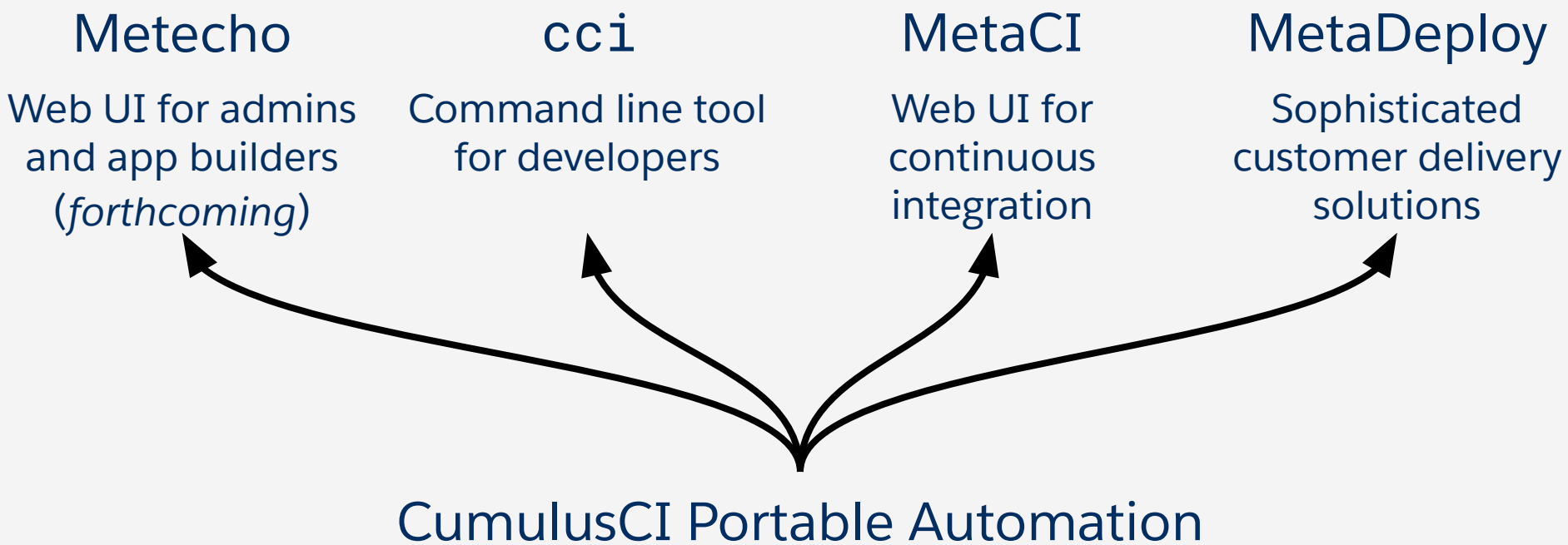


CumulusCI as the  
automation engine.





# Same automation, many contexts

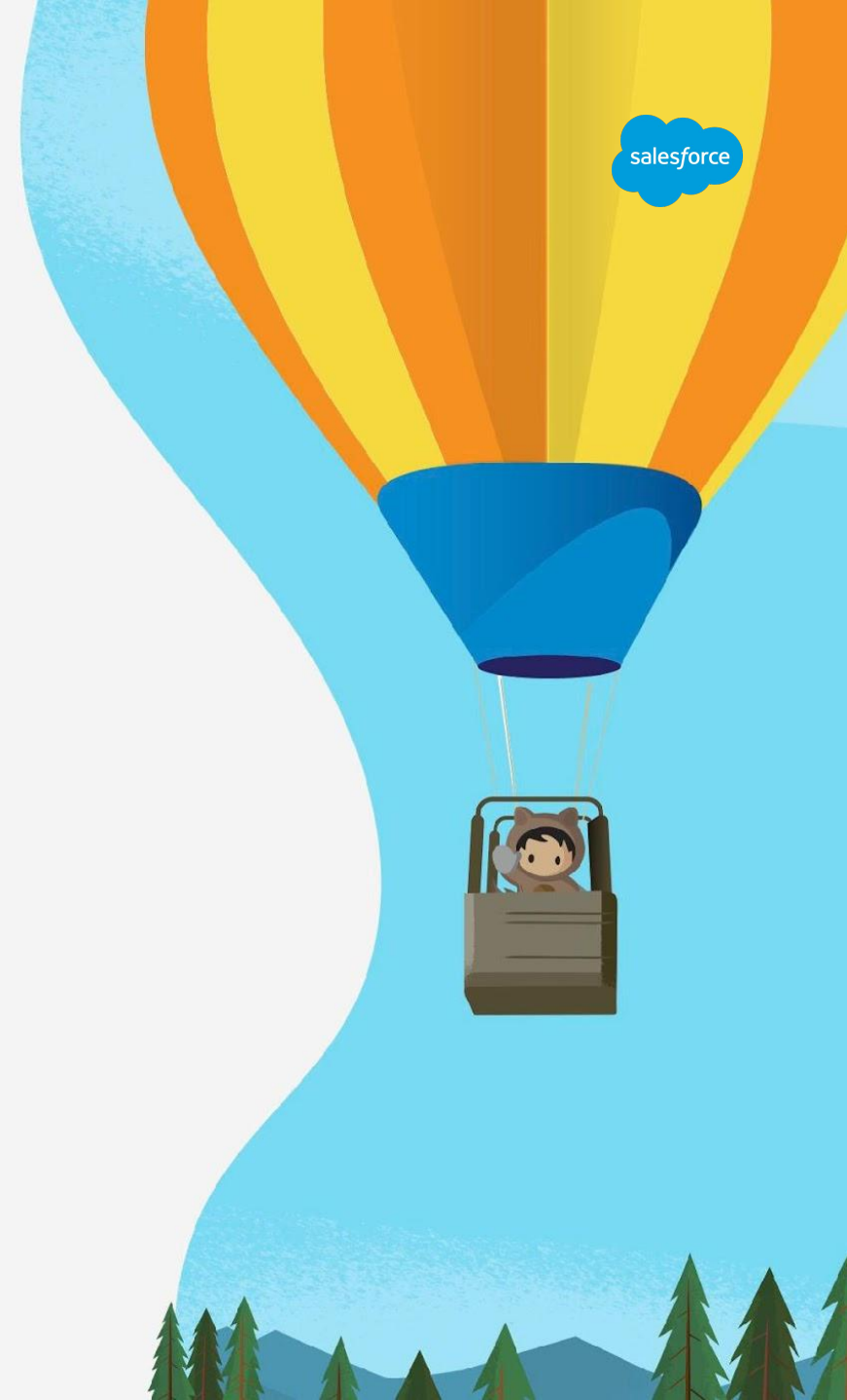


# What CumulusCI does

Builds orgs based on repeatable recipes

- Dependency management
- Package or application installation
- Metadata deployment to tailor org
- Configuration of settings
- Data seeding
- Any task achievable with the Salesforce API

CumulusCI makes it easy to define fully-realized scratch orgs for each of your roles.

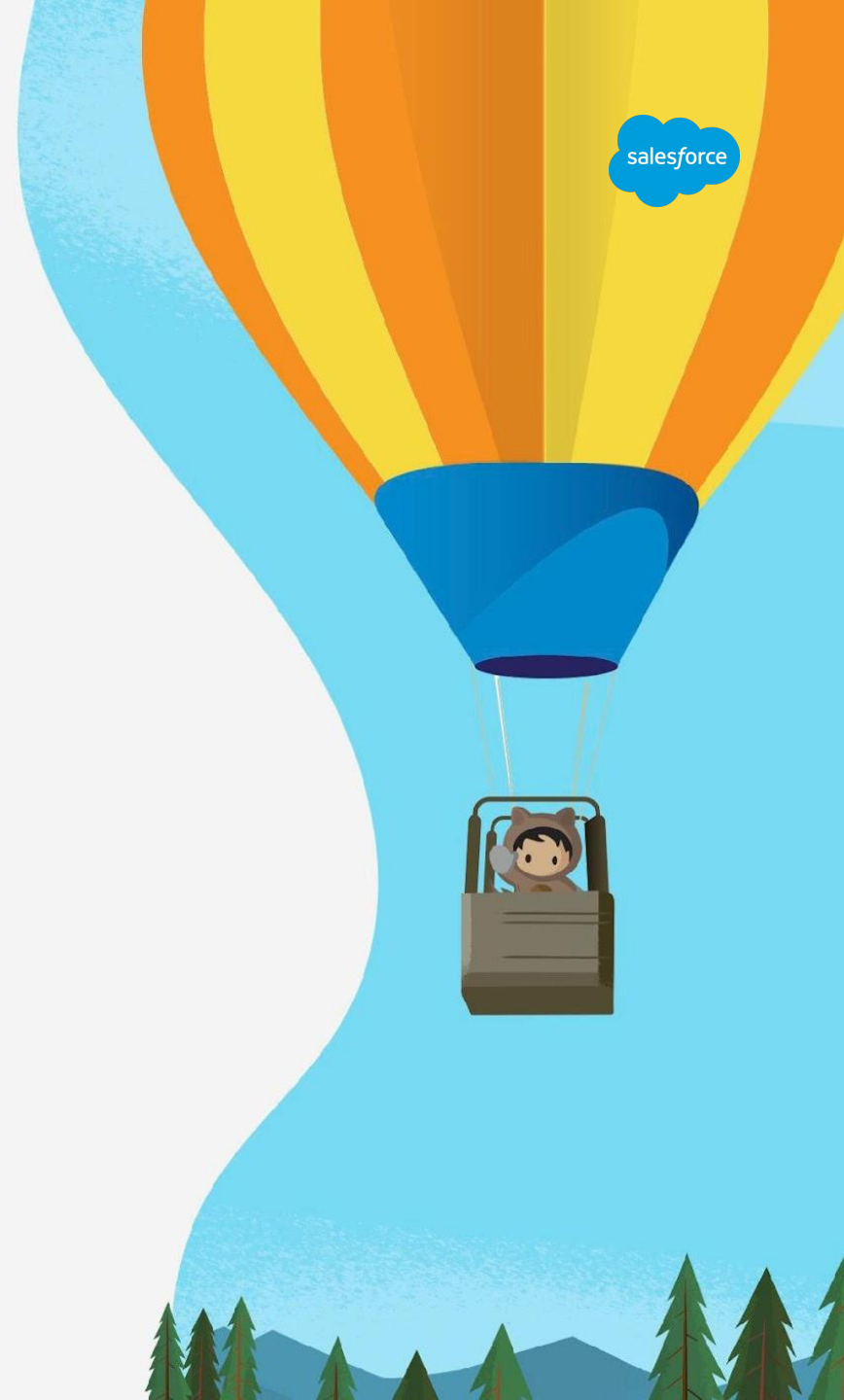


# What CumulusCI does

Provides tools for each role to use in their orgs

- Run tests (Apex, Jest, Robot browser and API automation)
- Develop and deploy code and metadata
- Load, modify, and capture data sets

CumulusCI makes it easy for developers, admins, and testers to do their work and collaborate in version control.

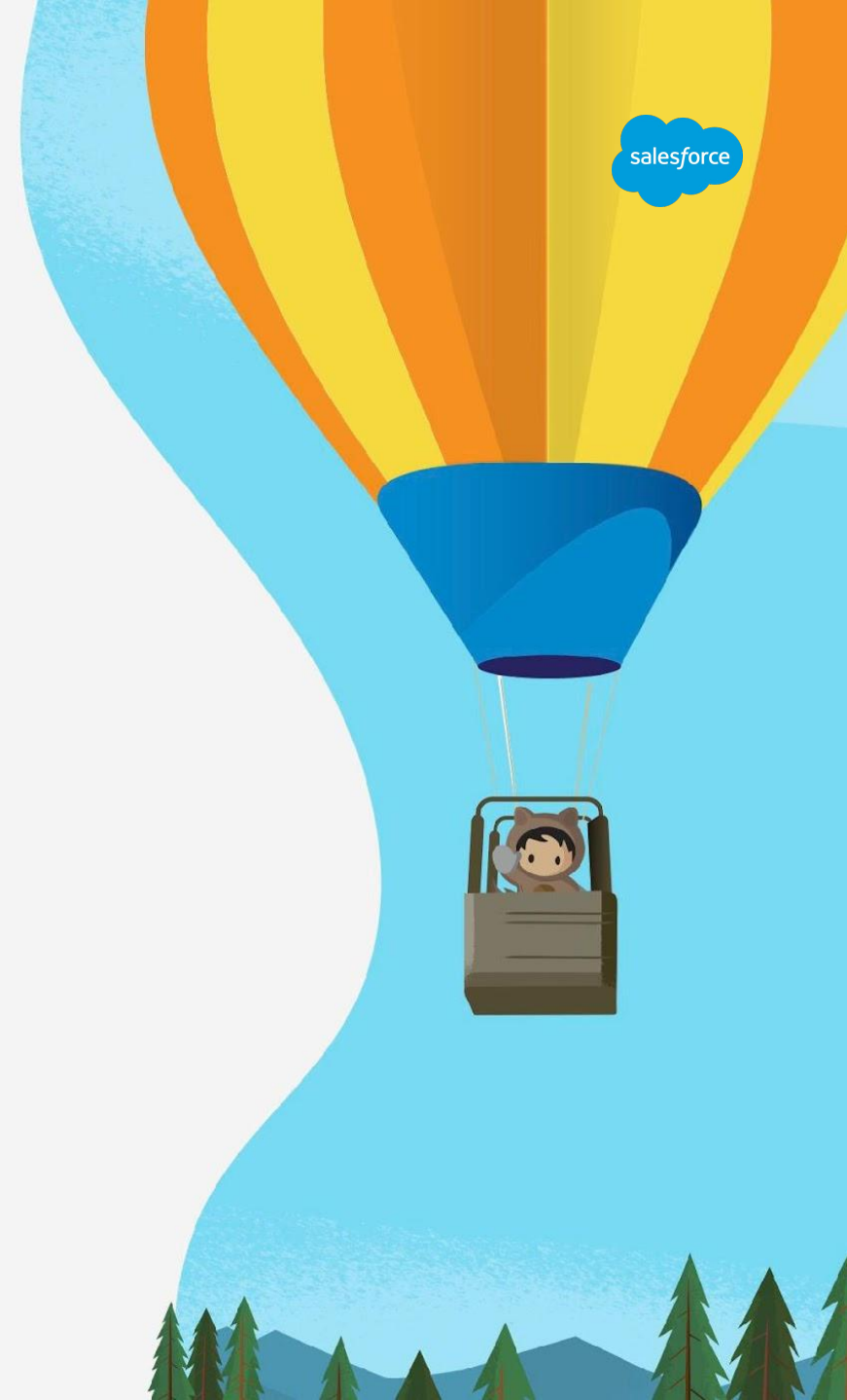


# What CumulusCI does

Provides tools to orchestrate the entire development process

- Run continuous integration builds
- Execute Apex, Jest, and Robot Framework browser tests
- Automate parent-child branch merges (CumulusCI Flow)
- Upload and test beta managed packages automatically
- Upload and test managed releases

Continuous integration server recommended but not required: you can run the entire development process from your local machine.



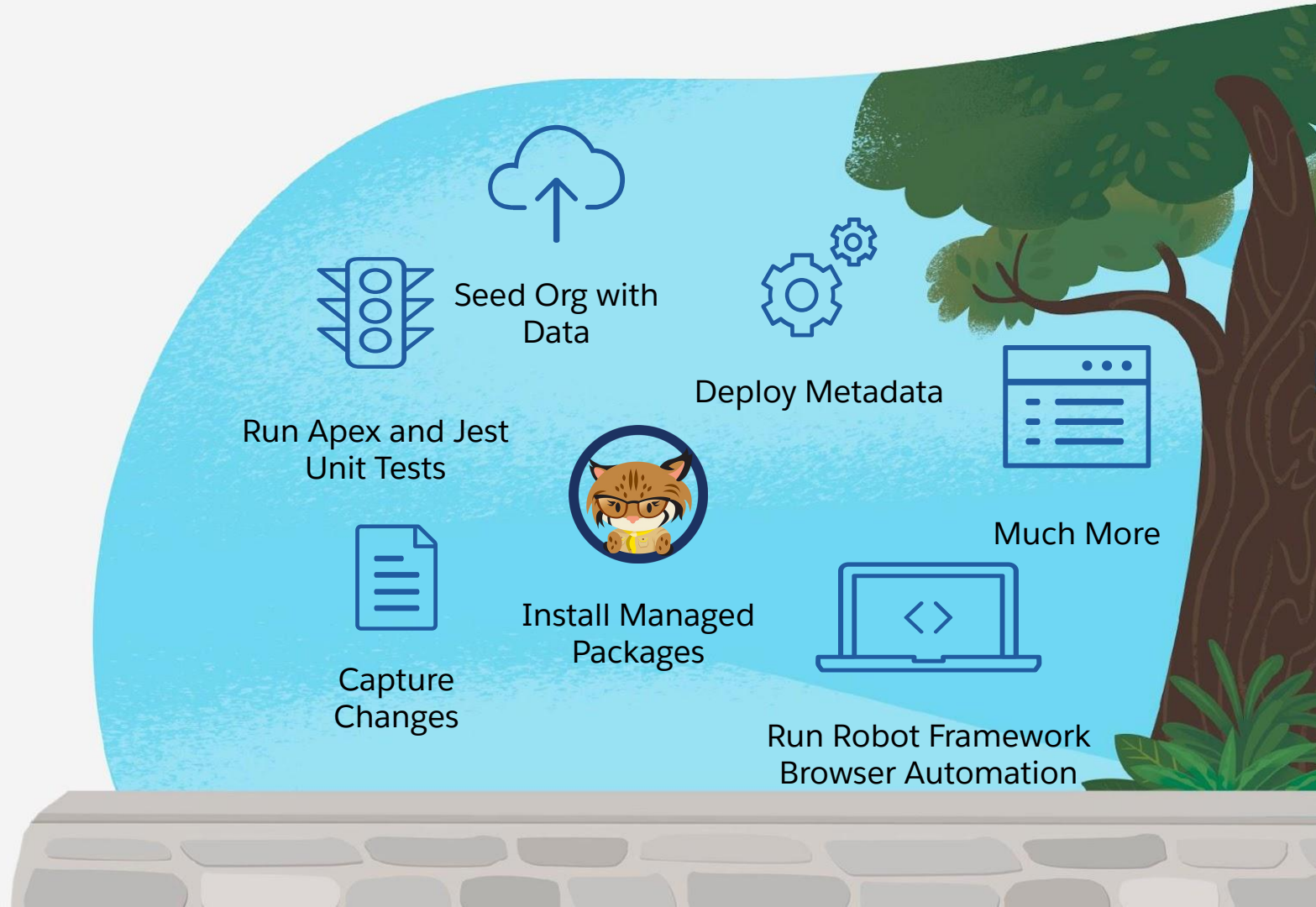
# Flows and tasks

## Composable building blocks for automation

CumulusCI comes with dozens of built-in tasks and predefined flows.

Tasks and flows are configurable in simple text markup.

New capabilities can be added in Python.





# Customize flows and tasks in version control

`cumulusci.yml` lives in the repository alongside code

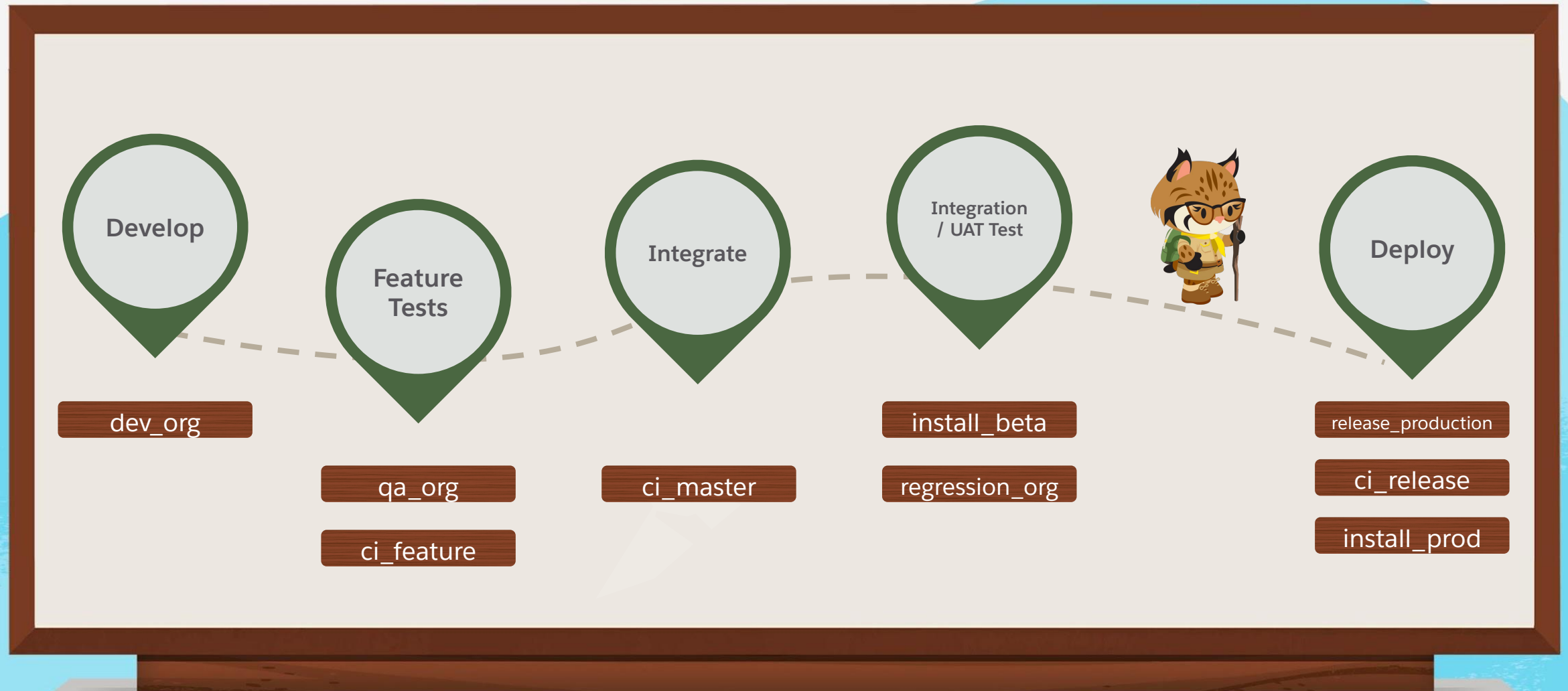
```
trailhead_org:
  steps:
    1:
      task: update_dependencies
    2:
      task: deploy
      options:
        path: unpackaged/pre/account_record_types
    3:
      task: ensure_record_type
    4:
      task: install_managed
    5:
      flow: config_trailhead_playground
```



A stylized illustration of a forest scene. Two large, leafy green trees with thick brown trunks frame the top and sides of the image. In the background, there are rolling green hills and several tall, thin evergreen trees. The sky is a light blue gradient. In the bottom right corner, there are some yellow flowers and green foliage. A small bird is perched on a branch of the tree on the left.

# Hands-On Demonstration

# The trail to production



# Outcomes of solving the org problem



## Increased velocity

- Portable automation means no org blockers or infrastructure dependencies

## Increased confidence

- Rely on orgs always being configured exactly as you expect

## Modern best practices not just baked-in, but enabled

- Source control
- Salesforce DX

## Users get access to holistic environments that encourage collaboration

- CumulusCI makes it easy to capture and share work.





# Metecho: Portable Automation for Admins

No command line? No problem.

*Metecho is not yet released.  
Functionality may change before release.*

*Μετέχω: “I share”, Ancient Greek*



# Metecho

Manage projects and tasks in your repositories



### Add Data Validation

/jlantz/CCI-Food-Bank

Home > CCI-Food-Bank > Add Data Validation

Planned

In progress

Review

Merged

Tasks for Add Data Validation

Add a Task

0 of 1 Complete

| Task                                            | Status                            | Developer | Reviewer |
|-------------------------------------------------|-----------------------------------|-----------|----------|
| <a href="#">Build Delivery Validation Rules</a> | <div><div></div>In Progress</div> |           |          |

# Metecho



Create scratch orgs, capture work into Git with the push of a button

## Build Delivery Validation Rules

[/jlantz/CCI-Food-Bank](#)

[Delete Task](#)[View Pull Request](#)

[Home](#) > [CCI-Food-Bank](#) > [Add Data Validation](#) > Build Delivery Validation Rules

### Task Team & Orgs

Developer

 davidmreed

Dev Org

Deployed Commit: [cc4fd56](#)  
Expires: in 7 days  
Status: up to date | [check again](#)

[View Org](#)

Reviewer

 jlantz

Review Org

Status: not yet created

### Build Delivery Validation Rules

### Commit History

[cc4fd56](#)

Add validation rule

about 1 hour ago

# Metecho Integrates with Git and Developers



Bring declarative work fully into your development lifecycle via Portable Automation

## Build Delivery Validation Rules #22

Open

davidmreed wants to merge 1 commit into `feature/add-data-validation` from `feature/add-data-validation`

Conversation 0 Commits 1 Checks 0 Files changed 3

davidmreed commented 2 hours ago 

Member

### Critical Changes

---

### Changes

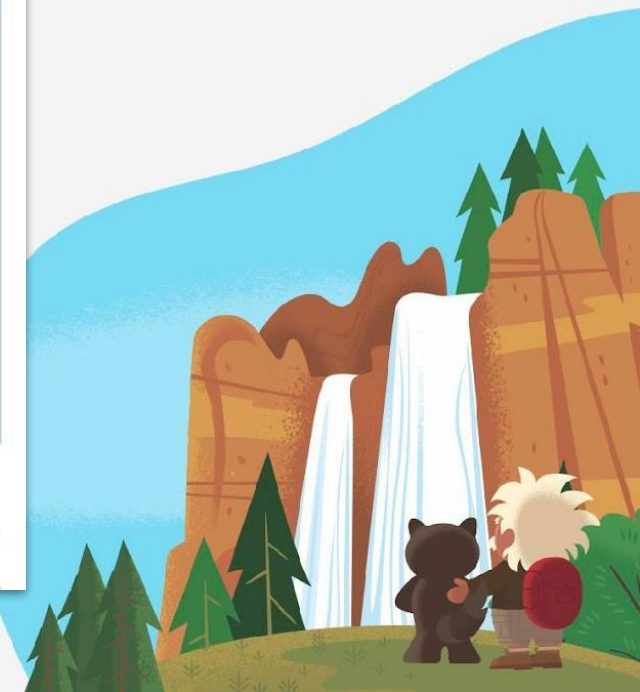
---

- Add validation rule on Delivery

### Issues Closed

---

Add validation rule cc4fd56





# Hands-On Demonstration

*Metecho is not yet released.  
Functionality may change before release.*



# Completing the Picture with MetaCI and MetaDeploy

Applying portable automation to continuous  
integration and customer delivery



# MetaCI



## Continuous integration for Salesforce projects

Orchestrate development workflow.

Easily define plans that run CumulusCI automation on specific branches and tags.

Status reported to GitHub for process insights.

Track releases and automatically generate release notes in GitHub.

 **All checks have passed** [Hide all checks](#)  
3 successful checks

|                                                                                    |                                                                                                                                                                                  |
|------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |  <b>Feature - Prerelease</b> — The build was successful <a href="#">Details</a>                 |
|  |  <b>Feature Test</b> — The build was successful <span>Required</span> <a href="#">Details</a> |
|  |  <b>Feature Test - Namespaced</b> — The build was successful <a href="#">Details</a>          |

Latest release

rel/3.117  
93f3eb9

Compare ▾

3.117

 salesforce-org-metaci released this 10 days ago

Changes



# MetaCI

Support declarative users with web UI

Create scratch orgs using CumulusCI automation through a web UI.



MetaCI

Create Org

Create Scratch Org

Use this section to create a scratch org from a branch for general feedback purposes.

Create Scratch Org

# Prefer to bring your own CI?

No lock-in



MetaCI makes it easy to map CumulusCI automation to your GitHub branches and tags and implement CumulusCI Flow.

CumulusCI works great in other CI systems too.

Use CumulusCI as your build tool in CircleCI, Jenkins, Travis, GitHub Actions, GitLab Pipelines, or whatever you prefer. Even on-premises!



# MetaDeploy

Customer delivery with portable automation



## Select a Product to Install



### Volunteers for Salesforce (V4S)

Version 3.116

Volunteer management for your organization.



### K-12 Architecture Kit

Version 1.5

A Custom Solution for K-12 Schools and Districts



### Nonprofit Success Pack (NPSP)

Version 3.169

The Foundation for the Connected Nonprofit



### Education Data Architecture (EDA)

Version 1.91

The Foundation for the Connected Campus



### Gift Entry Manager (GEM)

Version 1.10

Fundraising Tools for Your Institution





# MetaDeploy

Smart, product-oriented installers create a great customer experience



 Nonprofit Success Pack (NPSP), 3.171  
**Install NPSP**

Average Install Time: 14 minutes.

Welcome to the Nonprofit Success Pack installer!

**IMPORTANT:** Salesforce does not recommend installing directly in production organizations. Instead, install NPSP in one of your free sandbox environments or a developer organization first.

Log In to Install ▼

← [Select a different plan](#)



Not Connected to Salesforce

| Steps                                 | Type                                                                                                                        |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Install Contacts & Organizations 3.14 |  Managed Package <span>Required</span> |
| Install Households 3.12               |  Managed Package <span>Required</span> |
| Install Recurring Donations 3.17      |  Managed Package <span>Required</span> |
| Install Relationships 3.9             |  Managed Package <span>Required</span> |



CumulusCI makes sharing and collaboration easy – in one organization, across organizations, between roles, and in the open source community.







thank  
you

BLAZE  
YOUR  
TRAIL

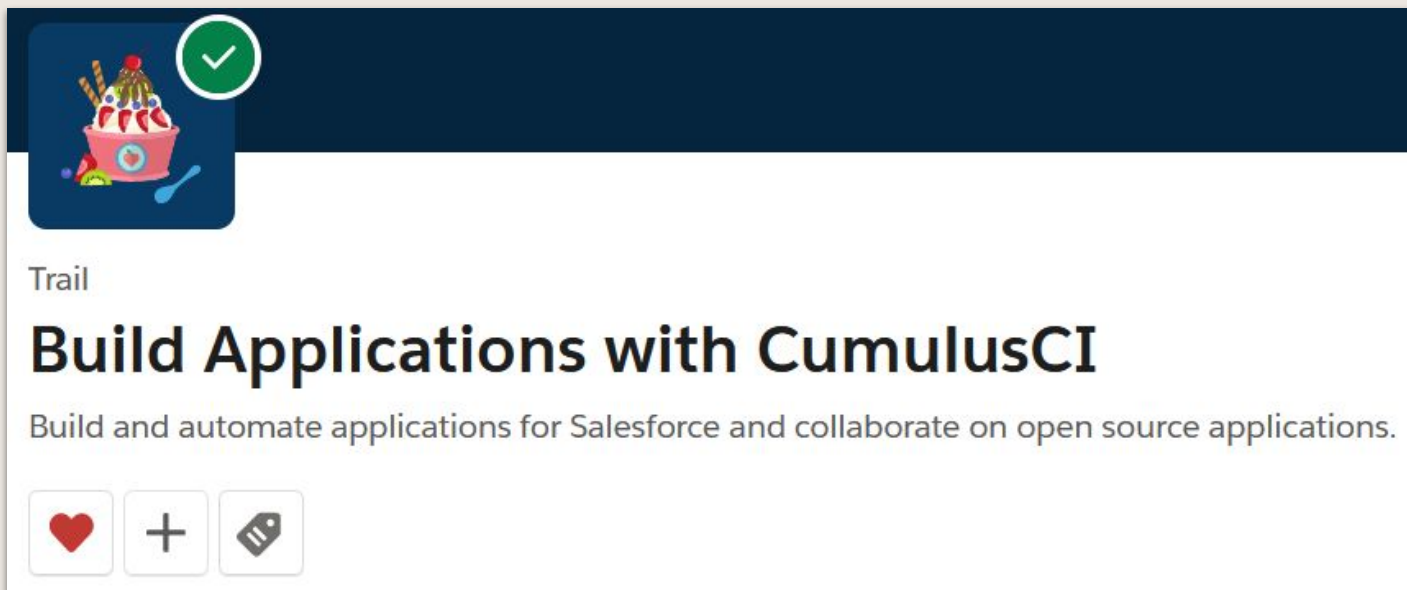
salesforce

# Now available on Trailhead!



## Build Applications with CumulusCI:

<https://trailhead.salesforce.com/en/content/learn/trails/build-applications-with-cumulusci>



The image shows a digital screen displaying a Trail card. The card has a dark blue header with a green checkmark icon in a white circle. Below the header is a white section with the word "Trail" in small text, followed by the title "Build Applications with CumulusCI" in large bold text. Under the title is a subtitle: "Build and automate applications for Salesforce and collaborate on open source applications." At the bottom of the card are three icons: a red heart, a plus sign, and a tag icon.

Trail

### Build Applications with CumulusCI

Build and automate applications for Salesforce and collaborate on open source applications.

♥ + 🏷️



# Resources



CumulusCI Documentation: [cumulusci.readthedocs.io](https://cumulusci.readthedocs.io)

CumulusCI on the Trailblazer Community:

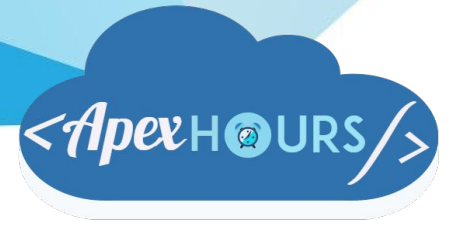
[https://success.salesforce.com/\\_ui/core/chatter/groups/GroupProfilePage?g=0F93000000009M9Z](https://success.salesforce.com/_ui/core/chatter/groups/GroupProfilePage?g=0F93000000009M9Z)

CumulusCI Open Source Project: [github.com/SFDO-Tooling/CumulusCI](https://github.com/SFDO-Tooling/CumulusCI)

Salesforce.org Products and the Open Source Commons

- [github.com/SalesforceFoundation/NPSP](https://github.com/SalesforceFoundation/NPSP)
- [github.com/SalesforceFoundation/EDA](https://github.com/SalesforceFoundation/EDA)
- [github.com/SFDO-Community](https://github.com/SFDO-Community)

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